

EXPLORING DENGUE IN SRI LANKA: A REPORT ON AN INTERACTIVE DENGUE SURVEILLANCE DASHBOARD



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Dashboard link

<https://nathashaninthushi-denguedashboard.share.connect.posit.cloud>

1. Introduction

Dengue is one of Sri Lanka's most significant public health challenges. It produces a heavy and recurring case burden that demands continuous surveillance to anticipate seasonal rises and to detect outbreaks early. This report accompanies an interactive dashboard, built in R and Quarto that turns the national weekly dengue surveillance record into an accessible exploratory tool supporting both high level summaries and district level investigation. The report explains the data, the analytical methods used with particular attention to trend and seasonality exploration and to automatic outbreak detection and the key findings, including the main observed patterns and outbreak periods. The dashboard itself is available at the link on the title page.

2. Data

The dashboard draws entirely on the **srilanka_weekly_data** dataset from the **denguedatahub** R package, which compiles the Weekly Epidemiological Reports of the Epidemiology Unit, Ministry of Health, Sri Lanka. The data are in long format, one row per reporting area per epidemiological week, recording the year, week number, the week's start and end dates, the reporting area, and the number of notified suspected dengue cases. In total the dataset holds **25,766 weekly observations across 26 reporting areas**, spanning 2006 to the latest reported week of December 2025, and amounting to **803,074** notified cases

One of the 26 reporting areas, **Kalmune** has no matching administrative boundary; it is therefore retained in all totals, rankings and time series but appears as an unmapped (“NA”) area on the choropleth map. Case counts are also not standardised by population, so densely populated districts naturally record larger totals.

3. Methods

3.1 Data preparation

All analysis was carried out in R. Data manipulation, date parsing and aggregation into national, annual and district totals used the tidyverse and lubridate. For time series analysis, epidemiological week labels were generated and the national series was placed on a regular weekly index (a tsibble) with any missing weeks filled as zero, enabling seasonal decomposition with feasts and fabletools, with xts supporting the interactive timeline.

3.2 Dashboard structure

The dashboard is organised into six pages. Overview presents the headline indicators together with the national weekly curve and annual totals; Districts provides a cumulative cases choropleth, a ranking table, and a weekly comparison in which any set of districts can be overlaid; By Year offers a district-by-year heatmap, small-multiple annual maps, and annual district trends; Seasonality & Trend shows the STL decomposition and a weekly heatmap; Outbreaks applies automatic anomaly detection and lists the flagged weeks; and Data provides a searchable browser of the raw records. A custom stylesheet gives the dashboard a dark navigation bar, clean value boxes and consistent colour accents.

3.3 Analytical and visualisation techniques

Interactive charts were built with plotly and dygraphs, spatial maps with leaflet, sf and the ceylon boundary package, and searchable tables with DT; viridis supplied perceptually uniform colour scales and crosstalk linked the district selectors to their charts. Trend and seasonality were separated using STL (Seasonal–Trend decomposition using Loess) with an annual 52-week period, which splits the series into a smooth trend, a repeating seasonal cycle and an irregular remainder. Outbreak periods were then identified by screening the remainder for statistical anomalies with an interquartile range rule, so that weeks lying far above the expected trend plus seasonal level are flagged automatically rather than against a fixed case threshold. To keep moderate districts legible despite the extreme skew of epidemic years, the maps and heatmaps use a square root colour transform.

4. Key Findings

4.1 Overall burden and the 2017 epidemic

Across the whole record the country reported **803,074** dengue cases, but the burden is strikingly uneven from year to year. The single dominant event is the **2017 epidemic**, which produced **174,687** cases, several times a typical year and drove national weekly incidence to a record **10,590 cases in the week ending 15 July 2017**. This surge is widely attributed to flooding following the 2017 southwest monsoon, which expanded mosquito breeding habitat. On every view of the dashboard the 2017 peak stands out as an extreme outlier, far above the next largest years.

4.2 Spatial distribution and district-level burden

Dengue is endemic nationwide, but its intensity is strongly concentrated in the Western Province. The district ranking is led by **Colombo (174,701 cumulative cases)** and **Gampaha (102,397)**, followed by Kandy (66,537), Kalutara (48,564) and Jaffna (44,908), a pattern the district-by-year heatmap confirms year after year, reflecting the dense, highly urbanised populations of the western districts. The picture is not purely western, however: Gampaha's most intense single week (2,631 cases) actually exceeded Colombo's (1,972), and the year-by-year maps show the 2017 epidemic intensifying across almost the entire island rather than the west alone.

4.3 Seasonality and long-term trend

The STL decomposition cleanly separates the disease's structure. The **trend** component shows that the underlying level was low and broadly stable until the mid 2010s, then sharply to a peak around 2017 (about 3,250 cases per week) before easing to lower, wavier with further humps around 2019–2020 and 2023–2024, the multi-year cycles typical of dengue. The **seasonal** component confirms a reliable annual rhythm whose amplitude has grown over time, from swings of a few hundred cases early in the record to well over a thousand in recent years; this widening is consistent with seasonal effects that scale with the overall level of transmission. Within each year the pattern is broadly **bimodal**, dominated by a mid year rise around weeks 20–31 (the southwest monsoon) with a smaller secondary rise late in the year (the northeast monsoon), a rhythm clearly visible as recurring vertical bands in the weekly heatmap.

4.4 Outbreak detection and outbreak periods

The Outbreaks page turns this into an explicit early warning view. Screening the remainder for anomalies flags clusters of unusually high weeks in three periods: **2017 (by far the largest), 2019–2020, and 2023–2024**. The flagged weeks are highly informative about timing. The ten most extreme outbreak weeks are almost all consecutive weeks of the 2017 mid-year epidemic, the four highest, each above 10,000 cases, fall in July 2017 (weeks ending 1, 8, 15 and 22 July) with the wider 2017 cluster spanning June to August. The largest flagged weeks outside 2017 occur in **mid-November 2019** (around 4,700–4,800 cases). The two biggest epidemics therefore peaked in different seasonal windows, 2017 in the mid-year southwest-monsoon season and 2019 late in the year during the northeast monsoon, showing that both of the country's rainfall seasons can drive major outbreaks.

5. Conclusion

The dashboard converts a large, weekly, spatially detailed surveillance dataset into an explorable picture of dengue in Sri Lanka, and three conclusions stand out. First, transmission is strongly and predictably seasonal, concentrated in a dominant mid-year period and a smaller late-year period tied to the two monsoons. Second, the burden is spatially concentrated in the Western Province, led by Colombo and Gampaha, so prevention resources directed at these high-density districts would reach the largest share of cases. Third, layered on top of this predictable structure are occasional anomalous epidemics, most dramatically 2017, then 2019, with a resurgence in 2023–2024, that far exceed seasonal expectation; the anomaly detection view distinguishes these genuine outbreaks from ordinary seasonal peaks and provides a simple early signal as new weeks arrive.